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ORIGINAL ARTICLE

# Analytical and numerical solutions of squeezing unsteady Cu and TiO<sub>2</sub>-nanofluid flow in the presence of thermal radiation and heat generation/absorption



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 Runge-Kutta scheme (RK4)

**Abstract** In this paper, an analytical study of squeezing unsteady nanofluid flow between two parallel plates is carried out. The effects of thermal radiation and heat generation/absorption along the temperature profile were observed. The homotopy perturbation method (HPM) was used to solve the two-dimensional nonlinear momentum and energy equations. The results found in this study are being compared with the result obtained using the RK4 technique with shooting scheme. The results obtained fascinatingly agree with the numerical solutions.

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## 1. Introduction

For decades, the study of flow and heat transfer for squeezing unsteady viscous fluid flow through equidistant plates has been an interesting research area for its vast scope of physical appliances. For instance, lubrication set up, food processing, hydrodynamical machines, compression, and damage of crops due to freezing, formation, and dispersion among others. Siddiqui et al. [1] studied and analyzed the unsteady two-dimensional flow of a viscous magnetohydrodynamic MHD fluid between

two parallel infinite plates. Their results show that, at a given time, and for a particular figure of the constant parameter  $R$ , the normal velocity rises significantly from the similarity variable  $\eta = 0$  to  $\eta = 1$ , for different values of the magnetic parameter  $M$ . Sheikholeslami et al. [2] evaluated the problem of squeezing unsteady flow of a nanofluid bounded by two parallel plates analytically. Their results display that as the plates are moving together, they are observed as both nanoparticle concentration and Eckert number incrementing, so does the Nusselt number. However, another interesting and related study is reported in [3]. Hamza [4] considered the motion of an electrically conducting fluid film squeezed between two parallel disks with a magnetic field. He discovered that the electromagnetic force increased the load carrying capacity significantly. The problem of squeezing flow between rotating

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**Nomenclature**

|                 |                                                |
|-----------------|------------------------------------------------|
| $A$             | general differential operator                  |
| $A_1, A_2, A_3$ | dimensionless constants                        |
| $B$             | boundary operator                              |
| $C_f$           | skin friction coefficient, defined by Eq. (16) |
| $Ec$            | Eckert number                                  |
| $f(\eta)$       | dimensionless function                         |
| $k_f$           | fluid's thermal conductivity (W/m K)           |
| $k_{nf}$        | thermal conductivity of the nanofluids         |
| $k_p$           | thermal conductivity of the solid material     |
| $l$             | initial position of the plates                 |
| $L$             | linear operator                                |
| $N$             | radiation parameter                            |
| $\mathcal{N}$   | nonlinear operator                             |
| $Nu$            | Nusselt number, defined by Eq. (16)            |
| $p$             | pressure                                       |
| $Pr$            | Prandtl number                                 |
| $q_r$           | radiative heat flux                            |
| $Q$             | heat generation/absorption coefficient         |
| $Re_x$          | local Reynolds number                          |
| $S$             | squeezing integer                              |
| $t$             | dimensionless time                             |
| $T$             | local fluid temperature (K)                    |
| $T_\infty$      | surrounding temperature                        |
| $u, v$          | velocity components (m/s)                      |
| $x, y$          | axial coordinates (m)                          |

*Greek symbols*

|              |                                              |
|--------------|----------------------------------------------|
| $\alpha$     | is a constant                                |
| $\eta$       | similarity variable                          |
| $\phi$       | dimensionless nanoparticle volume fraction   |
| $\Lambda$    | boundary                                     |
| $\lambda$    | heat generation parameter                    |
| $\mu_f$      | fluid's dynamic viscosity (kg/m s)           |
| $\mu_{nf}$   | dynamic viscosity of the base fluid (kg/m s) |
| $\Omega$     | domain of the boundary                       |
| $\rho_f$     | fluid's density (kg/m <sup>3</sup> )         |
| $(\rho C)_p$ | heat capacity of the nanoparticle material   |
| $\rho_{nf}$  | effective density of the nanofluid           |
| $\rho_p$     | nanoparticle mass density                    |
| $\theta$     | dimensionless temperature                    |

*Superscripts*

|     |                                        |
|-----|----------------------------------------|
| $'$ | differentiation with respect to $\eta$ |
|-----|----------------------------------------|

*Subscripts*

|          |                              |
|----------|------------------------------|
| $f$      | fluid                        |
| $nf$     | nanofluid                    |
| $p$      | particle                     |
| $\infty$ | condition at the free stream |

disks has been studied by [5] and later by Bhattacharyya and Pal [6]. He [7–10] introduced a new perturbation technique which is indeed a pairing of the classical perturbation method as Nayfeh [11], Krylovas [12], and the homotopy used in topology. This idea makes the HPM.

Recently, Dib et al. [13] studied and approximated an analytical solution of squeezing unsteady nanofluid flow by using Duan-Rach approach (DRA). They obtained a good agreement with the numerical method by using fourth order Runge-Kutta algorithm. Furthermore, Pourmehran et al. [14] studied the problem of squeezing unsteady nanofluid flow between two equidistant plates by least square method (LSM) and collocation method (CM). Their results indicate that the Nusselt number increases with respect to an increment from both the nanoparticle volume fraction and the Eckert number, while it reduces with an increase in the squeezing number. There are some essential studies in relation to the squeezing fluid flow problem [15–19].

Radiation is a heat transfer technique which does not depend on any contact between the heat source and the heated object, unlike the conduction and convection cases where they paramountly needed some contact. Heat can be conveyed through an unoccupied gap by thermal radiation mostly termed as infrared radiation. Kandasamy et al. [20] investigated the problem of unsteady Hiemenz flow of Cu-nanofluid over a porous wedge with thermal stratification caused by solar energy radiation. Moreover, some very attractive analyses on the fluid flow with thermal radiation effects have been made and reported in [21–24].

The study of heat generation or absorption effects is paramount in cooling processes. Therefore, there are many published studies related to the heat generation cases. The heat generation/absorption effect on stagnation point flow of nanofluid over a surface is analyzed by Alsaedi et al. [25]. They used a homotopy analysis method (HAM) in the analysis, and they found that heat source leads to increase in the fluid temperature. Mansour and Ahmed [26] investigated the heat generation effects on natural convective heat transfer in an inclined triangular enclosure filled with a Cu-water nanofluid saturated porous medium using an implicit finite difference method. They observed that a decent growth in the average Nusselt number could be achieved by an increase in the nanoparticle volume fraction. Besides, the average Nusselt number decreases as the heat source parameter increases. Furthermore, the influence of internal heat generation/absorption on magnetohydrodynamic natural convection from an isothermal truncated cone was numerically studied by Srinivasa and Eswara [27], in which they discover that the Nusselt number decreases due to the increase in the heat generation/absorption parameter value. Abdel-wahed et al. [28] consider the heat generation effects on the flow and heat transfer of nanofluid over a moving surface using the optimal homotopy asymptotic method (OHAM). Their analysis on the heat source parameter  $\lambda$  shows that the boundary layer temperature relatively increases as  $\lambda$  is growing, while the nanoparticle concentration near the surface reduces as  $\lambda$  is increasing. Wahiduzzaman et al. [29] studied the influence of thermal radiation, heat generation and chemical reaction on MHD Convective stagnation flow of nanofluid

over a shrinking surface using a six-order Runge-Kutta method with shooting technique. Ultimately, another recent and interesting study concerning the influence of heat generation/absorption on the flow and heat transfer problem is reported in [30]. More and recent studies concerning the role being played by nanofluid during the heat transfer process can be found in [31–34].

The purpose of the present study was to use HPM to derive the solution of squeezing unsteady nanofluid flow analytically. The thermal radiation and heat generation/absorption effects on the temperature profile were analyzed. Besides, the fourth-order Runge-Kutta (RK4) method along with shooting technique was used for the authentication of our results. On the other hand, when considering both the thermal radiation and heat generation/absorption terms being limited to zero, then our result would certainly diminish to the past published results found by [2,3]. All the salient parameters associated with this study were duly analyzed.

## 2. Problem description

In the present study, the flow and heat transfer of two-dimensional unsteady nanofluid squeezed through the equidistant plates was observed as depicted in Fig. 1. The distance between the two plates at any non-dimensional time  $t$  is given as  $w = \pm l(1 - \alpha t)^{\frac{1}{2}} = \pm h(t)$ . For  $\alpha > 0$ , the two plates are squeezed until they touch each other at  $t = \frac{1}{\alpha}$ . While for  $\alpha < 0$ , the two plates are separated, where  $\alpha$  is a constant and  $l$  is the initial position (at  $t = 0$ ) and  $z$  is the axial coordinate which is obviously considered as zero from the flow region, with the flow model as considered along  $x$  and  $y$  coordinates respectively. The variable  $t$ , is the non-dimensional time throughout the flow. The viscous dissipation effect and heat source as result of friction originated from the fluid flow shear are secured. Thus, this conduct befalls when the Eckert number is fantastically large. Meanwhile, the fluid is nanofluid which contained both copper and titanium oxide. The governing equations are as follows:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \quad (1)$$

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho_{nf}} \frac{\partial p}{\partial x} + \frac{\mu_{nf}}{\rho_{nf}} \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \quad (2)$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho_{nf}} \frac{\partial p}{\partial y} + \frac{\mu_{nf}}{\rho_{nf}} \left( \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right), \quad (3)$$

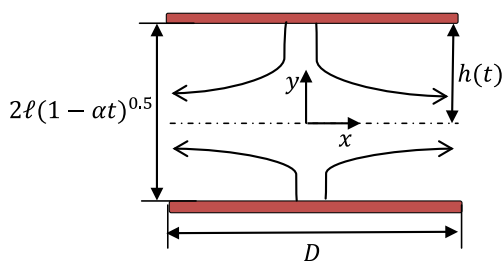


Figure 1 Geometry configuration of the problem.

$$\begin{aligned} \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = & \frac{k_{nf}}{(\rho C_p)_{nf}} \left( \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \\ & + \frac{\mu_{nf}}{(\rho C_p)_{nf}} \left[ 4 \left( \frac{\partial u}{\partial x} \right)^2 + \left( \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \right)^2 \right] \\ & - \frac{1}{\rho C_p} \frac{\partial q_r}{\partial y} + \frac{Q(T - T_\infty)}{\rho_{nf}}, \end{aligned} \quad (4)$$

Here  $u$  and  $v$  represent the velocities in  $x$  and  $y$  directions, respectively. While  $p$ ,  $T$ ,  $T_\infty$ ,  $f$ ,  $\rho$ , and  $Q$  are the pressure, the fluid temperature, the surrounding temperature, the fluid, the density and the heat generation/absorption coefficient, respectively, and  $\rho_{nf}$ ,  $\mu_{nf}$ ,  $(\rho C_p)_{nf}$  and  $k_{nf}$  are correspondingly the effective density, dynamic viscosity, heat capacity and thermal conductivity of the nanofluid as described in [35]:

$$\rho_{nf} = (1 - \phi)\rho_f + \phi\rho_p,$$

$$(\rho C_p)_{nf} = (1 - \phi)(\rho C_p)_f + \phi(\rho C_p)_p,$$

$$\mu_{nf} = \frac{\mu_f}{(1 - \phi)^{2.5}}, \quad (\text{Brinkman})$$

$$k_{eff} = \phi k_p + (1 - \phi)k_f, \quad (\text{Bhattacharya et al. [36]}) \quad (5)$$

where  $k_f$  is the thermal conductivity of the base fluid,  $k_p$  is the thermal conductivity of the nanoparticles,  $k_{eff}$  is the effective thermal conductivity of the nanofluid, and  $\phi$  is the nanoparticle volume fraction respectively.

With boundary conditions:

$$v = v_w = \frac{dh}{dt}, \quad T = T_H \quad \text{at } y = h(t),$$

$$v = \frac{\partial u}{\partial y} = \frac{\partial T}{\partial y} = 0 \quad \text{at } y = 0. \quad (6)$$

The radiative heat flux term in Eq. (4) was explicitly discussed by Rosseland [37] such that:

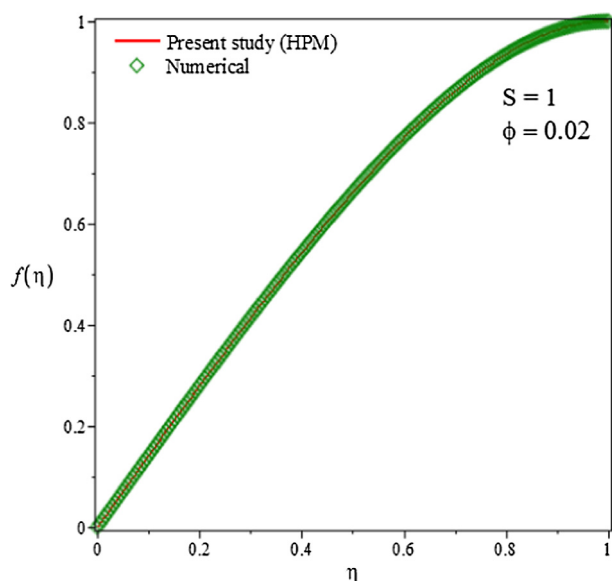
$$q_r = -\frac{4\sigma^*}{3k^*} \frac{\partial T^4}{\partial y}, \quad (7)$$

where  $\sigma^*$  and  $k^*$  are the Stefan-Boltzmann constant and the mean absorption number. Regarding some studies [38] and [39], we presume that the temperature variation amidst the flow is significantly limited, and the expression  $T^4$  may be considered as a linear function of temperature. Therefore,  $T^4$  is expanded using Taylor series expansion about  $T_\infty$  and ignoring the higher-order terms, gives

$$T^4 = 4T_\infty^3 T - 3T_\infty^4. \quad (8)$$

Substituting Eqs. (7) and (8) into Eq. (4) we obtain

$$\begin{aligned} \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = & \frac{K_{nf}}{(\rho C_p)_{nf}} \left( \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \\ & + \frac{\mu_{nf}}{(\rho C_p)_{nf}} \left[ 4 \left( \frac{\partial u}{\partial x} \right)^2 + \left( \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \right)^2 \right] \\ & + \frac{32\sigma^* T_\infty^3}{3\rho C_p k^*} \frac{\partial^2 T}{\partial y^2} + \frac{Q(T_w - T_\infty)}{\rho_{nf}}. \end{aligned} \quad (9)$$



**Figure 2** Comparison between the present study and numerical solutions (TiO<sub>2</sub>-water).

Introducing the following quantities,

$$\eta = \frac{y}{[l(1-\alpha t)^{\frac{1}{S}}]}, \quad u = \frac{\alpha x}{[2(1-\alpha t)]} f'(\eta),$$

$$v = -\frac{\alpha l}{[2(1-\alpha t)]} f(\eta),$$

$$\theta(\eta) = \frac{T}{T_H}, \quad A_1 = (1-\phi) + \phi \frac{\rho_s}{\rho_f}, \quad N = \frac{4\sigma^* T_\infty^3}{k_f \rho C_p k^*},$$

$$\lambda = \frac{Q2l(T_w - T_\infty)(1-\alpha t)}{K_f \rho_f T_w} \quad (10)$$

and substituting the non-dimensional variables in Eq. (10) into Eqs. (2)–(4) and eliminating the pressure gradient from the resulting equations (2) and (3) we obtain:

$$f^{iv} - SA_1(1-\phi)^{2.5}(\eta f''' + 3f'' + f'f'' - ff''') = 0 \quad (11)$$

$$(12A_3 + 16A_2N)\theta'' + 3PrSA_2(f\theta' - \eta\theta') + \frac{3PrEc}{(1-\phi)^{2.5}}(f''^2 + 4\delta^2 f'^2) + 3\lambda A_2\theta = 0 \quad (12)$$

where  $N$  is the thermal radiation,  $\lambda$  is the heat generation parameter given in Eq. (10), while  $A_2$  and  $A_3$  are dimensionless constants given by the following:

$$A_2 = (1-\phi) + \phi \frac{(\rho C_p)_s}{(\rho C_p)_f}, \quad A_3 = \frac{k_{nf}}{k_f}. \quad (13)$$

Now Eqs. (11) and (12) need to be solved subject to Eq. (14):

$$f(0) = 0, \quad f''(0) = 0, \quad f(1) = 1, \quad f'(1) = 0, \\ \theta'(0) = 0, \quad \theta(1) = 1. \quad (14)$$

Here  $S$  is the squeeze integer,  $Pr$  and  $Ec$  are the Prandtl and Eckert numbers, respectively. See Mustafa et al. [40]:

$$S = \frac{\alpha l^2}{2v_f}, \quad Pr = \frac{\mu_f(\rho C_p)_f}{\rho_f k_f},$$

$$Ec = \frac{\rho_f}{(\rho C_p)_f} \left( \frac{\alpha x}{2(1-\alpha t)} \right)^2, \quad \delta = \frac{l}{x}. \quad (15)$$

The following quantities are categorically used as defined in Domairry and Hatami [42]:

$$C_f = \frac{\mu_{nf} \left( \frac{\partial u}{\partial y} \right)_{y=h(t)}}{\rho_{nf} v_w^2}, \quad Nu = \frac{-lk_{nf} \left( \frac{\partial T}{\partial y} \right)_{y=h(t)}}{k T_H}. \quad (16)$$

It can be obtained from Eq. (10) that

$$C_f^* = \frac{l^2}{x^2(1-\alpha t)Re_x C_f} = A_1(1-\phi)^{2.5} f''(1),$$

$$Nu^* = \sqrt{1-\alpha t} Nu = -A_3 \theta'(1). \quad (17)$$

### 3. Homotopy perturbation technique

To illuminate the fundamental concept of the technique, as given by He [7,8,41] we take the general form of nonlinear differential equation into consideration as follows:

$$A(u) - g(r) = 0, \quad r \in \Omega, \quad (18)$$

alongside the boundary conditions

**Table 1** Comparison between the numerical and present study for  $f(\eta)$  and  $\theta(\eta)$  when  $S = 1$ ,  $Pr = 6.20$ ,  $Ec = 0.010$ ,  $\phi = 0.020$  (Cu-water),  $\delta = 0.010$  and  $N = \lambda = 0$ .

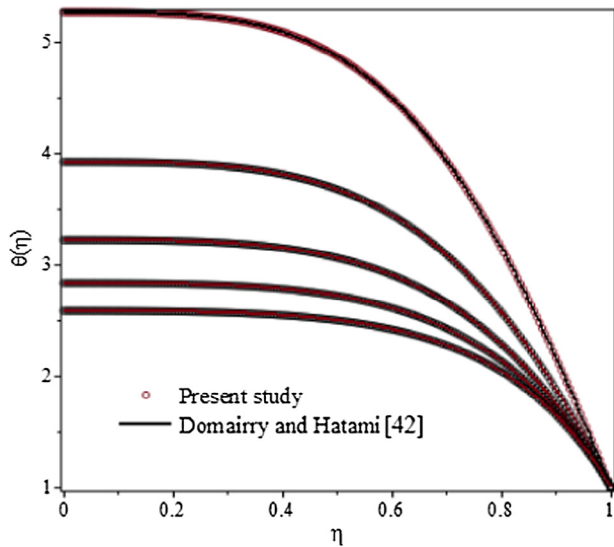
| $\eta$ | $f(\eta)$       |                     | $\theta(\eta)$  |                     |
|--------|-----------------|---------------------|-----------------|---------------------|
|        | Numerical (RK4) | Present study (HPM) | Numerical (RK4) | Present study (HPM) |
| 0      | 0.000000        | 0.000000            | 1.0320666       | 1.0321930           |
| 0.1    | 0.141359        | 0.141030            | 1.0320643       | 1.0321734           |
| 0.2    | 0.280666        | 0.280090            | 1.0320322       | 1.0321441           |
| 0.3    | 0.415781        | 0.415034            | 1.0318928       | 1.0319351           |
| 0.4    | 0.544379        | 0.543767            | 1.0315083       | 1.0313706           |
| 0.5    | 0.663857        | 0.663430            | 1.0306644       | 1.0301837           |
| 0.6    | 0.771229        | 0.771052            | 1.0290400       | 1.0289811           |
| 0.7    | 0.863016        | 0.863061            | 1.0261522       | 1.0260653           |
| 0.8    | 0.935120        | 0.935264            | 1.0212593       | 1.0211084           |
| 0.9    | 0.982695        | 0.982786            | 1.0131862       | 1.0130714           |
| 1      | 1               | 1                   | 1               | 1                   |

**Table 2** Comparison for the results of  $-\theta'(1)$  between the present study and previous study by Sheikholeslami et al. [2] when  $S = 0.5$  and  $\delta = 0.1$ .

| $Pr$ | $Ec$ | Sheikholeslami et al. [2] | Present study (HPM) |
|------|------|---------------------------|---------------------|
| 0.5  | 1    | 1.522367                  | 1.5223675           |
| 1    | 1    | 3.026324                  | 3.0263236           |
| 2    | 1    | 5.980530                  | 5.9805304           |
| 5    | 1    | 14.43941                  | 14.439413           |
| 1    | 0.5  | 1.513162                  | 1.5131618           |
| 1    | 1.2  | 3.631588                  | 3.6315883           |
| 1    | 2    | 6.052647                  | 6.0526471           |
| 1    | 5    | 15.13162                  | 15.131618           |

**Table 3** Effect of  $\lambda$  on  $-\theta'(1)$  when  $S = 1$ ,  $Ec = 0.5$ ,  $Pr = 6.2$ ,  $N = 0.5$ ,  $\phi = 0.02$  and  $\delta = 0.01$ .

| $\lambda$     | 0.0    | 0.5    | 1      | 3      | 50     |
|---------------|--------|--------|--------|--------|--------|
| $-\theta'(1)$ | 1.3998 | 1.4711 | 1.5424 | 1.8275 | 8.5288 |

**Figure 3** Validation of the results for  $\theta(\eta)$  (Cu-water) between the present study and previous results by Domairry and Hatami [42].

$$B\left(u, \frac{\partial u}{\partial n}\right) = 0, \quad r \in \Lambda, \quad (19)$$

in which  $A$  is the general differential operator,  $u(r)$  is the fluid velocity,  $B$  is the boundary operator, and  $g(r)$  is the known analytic function, with  $\Lambda$  being the boundary of the domain  $\Omega$ .  $A$  is the operator, which can be divided into both linear,  $L$  and nonlinear,  $N$ , parts, respectively. Hence Eq. (18) could be defined in the following form:

$$L(u) + N(u) - g(r) = 0. \quad (20)$$

Using the homotopy technique, the homotopy  $v(r, p) : \Omega \times [0, 1] \rightarrow \mathbb{R}$  is constructed, and it suffices the following expression:

**Table 4** Effect of  $Pr$  and  $Ec$  on  $-\theta'(1)$  when  $S = 1$ ,  $N = 1.5$ ,  $\phi = 0.02$ ,  $\lambda = 0.5$  and  $\delta = 0.01$ .

| $Pr$          | 0.5     | 1       | 1.5     | 3       | 1.5     |
|---------------|---------|---------|---------|---------|---------|
| $Ec$          | 1.5     | 1.5     | 1.5     | 1.5     | 0.5     |
| $-\theta'(1)$ | 0.23287 | 0.42525 | 0.61762 | 1.19474 | 0.23287 |

$$H(v, p) = (1 - p)[L(v) - L(u_0)] + p[A(v) - g(r)], \quad (21)$$

where  $p \in [0, 1]$  is called an embedding parameter, while  $u_0$  is the initial approximation of Eq. (18). Therefore, we can also write these as follows:

$$H(v, 0) = L(v) - L(u_0),$$

$$H(v, 1) = A(v) - g(r). \quad (22)$$

Consider the solution of equation (21) to be in the form of power series in  $p$  as follows:

$$v = v_0 + pv_1 + p^2v_2 + \dots, \quad (23)$$

where the final solution can be obtained when:

$$u = \lim_{p \rightarrow 1} v = v_0 + v_1 + v_2 + \dots. \quad (24)$$

#### 4. Solution with homotopy perturbation method

Here, we apply the method to solve the velocity profile in the momentum equations (2) and (3). Then the value of the velocity obtained is used to solve the energy equation (4) so as to find the temperature distribution considering the effect of heat generation/absorption. According to Eq. (20), the differential equations (11) and (12) are decomposed into linear  $L$  and nonlinear  $N$  operators, where  $L(u) = f^{iv}$ ,  $N(u) = SA_1(1 - \phi)^{2.5}(\eta f''' + 3f'' + f'f'' - ff''')$  and  $g(r) = 0$ . While  $L(u) = (12A_3 + 16A_2N)\theta''$  and  $N(u) = 3PrSA_2(f\theta' - \eta\theta') + \frac{3PrEc}{(1-\phi)^{2.5}}(f'^2 + 4\delta^2f'^2) + 3\lambda A_2\theta$  respectively.

We now apply HPM which satisfies Eq. (21) to nonlinear ordinary differential equation Eq. (11) as follows:

$$(1 - p)[f^{iv}] + p[f^{iv} + SA_1(1 - \phi)^{2.5}(\eta f''' + 3f'' + f'f'' - ff''')] = 0, \quad p \in [0, 1] \quad (25)$$

Assuming the solution of Eq. (25) can be written in following form

$$f = f_0 + pf_1 + p^2f_2 + \dots, \quad (26)$$

substituting Eq. (26) into Eq. (25) and using the boundary conditions (14), after some simplification and associating the like powers of  $p$ -terms, we obtain the following:

$$p^0 : f_0^{iv} = 0, \quad (27)$$

$$f_0(0) = 0, \quad f_0''(0) = 0, \quad f_0(1) = 1, \quad f_0'(1) = 0, \quad (28)$$

$$p^1 : f_1^{iv} + SA_1(1 - \phi)^{2.5}(\eta f_0''' + 3f_0'' + f_0'f_0'' - f_0f_0''') = 0, \quad (29)$$

$$f_1(0) = 0, \quad f_1''(0) = 0, \quad f_1(1) = 0, \quad f_1'(1) = 0. \quad (30)$$

Solving Eqs. (27)–(30), we obtained the following approximations:

$$f_0(\eta) = \frac{1}{2}(3\eta - \eta^3), \quad (31)$$

**Table 5** Effects of nanoparticle volume fraction and squeeze integer on skinfriction and Nusselt number, when  $Ec = 1, Pr = 1.5, N = 0.5, \lambda = 1.5$  and  $\delta = 0.01$ .

| $\phi$ | $S$ | $f''(1)$ | $\theta'(1)$ | $C_{fx}$ | $Nu$    |
|--------|-----|----------|--------------|----------|---------|
| 0.0    | 0.0 | -3.00000 | -0.85783     | -3.53719 | 0.92898 |
| 0.01   | 0.5 | -3.41065 | -0.87422     | -3.92159 | 0.94672 |
| 0.03   | 1.0 | -3.78044 | -0.90879     | -4.13056 | 0.98416 |
| 0.05   | 3.0 | -5.22249 | -0.94594     | -5.41656 | 1.02439 |

$$f_1(\eta) = \begin{cases} -0.0928571S\eta\sqrt{1-\phi}A_1 + 0.189286S\eta^3\sqrt{1-\phi}A_1 \\ -0.1S\eta^5(1-\phi)^{2.5}A_1 + 0.00357143S\eta^7(1-\phi)^{2.5}A_1 \\ +0.185714S\eta\sqrt{1-\phi}\phi A_1 - 0.378571S\eta^3\sqrt{1-\phi}\phi A_1 \\ -0.0928571S\eta\sqrt{1-\phi}\phi^2 A_1 + 0.189286S\eta^3\sqrt{1-\phi}\phi^2 A_1. \end{cases} \quad (32)$$

Hence, the required solution of Eq. (11), subject to Eq. (14) is obtained using the expression (24) as follows:

$$f(\eta) = f_0(\eta) + f_1(\eta) + \dots \quad (33)$$

where  $f_0$  and  $f_1$  are given by expression (31) and (32) respectively.

Similarly, having applied the same process in Eqs. (12) and (14), we obtained:

$$\theta_0(\eta) = 1, \quad (34)$$

$$\theta_1(\eta) = \begin{cases} \frac{1}{(12A_3+16A_2N)(1-\phi)^5} (-2.25EcPr\eta^4\sqrt{1-\phi} \\ -13.5EcPr\eta^2\delta^2\sqrt{1-\phi} + 4.5EcPr\eta^4\delta^2\sqrt{1-\phi} \\ -0.9EcPr\eta^6\delta^2\sqrt{1-\phi} + 2.25EcPr(1-\phi)^{2.5} \\ +9.9EcPr\delta^2(1-\phi)^{2.5} + 1.5A_2\lambda(1-\phi)^3 \\ -1.5A_2\lambda\eta^2(1-\phi)^3 + 4.5EcPr\eta^4\sqrt{1-\phi}\phi \\ +27EcPr\eta^2\delta^2\sqrt{1-\phi}\phi - 9EcPr\eta^4\delta^2\sqrt{1-\phi}\phi \\ +1.8EcPr\eta^6\delta^2\sqrt{1-\phi}\phi - 3A_2\lambda(1-\phi)^3\phi \\ +3A_2\lambda\eta^2(1-\phi)^3\phi - 2.5EcPr\eta^4\sqrt{1-\phi}\phi^2 \\ -13.5EcPr\eta^2\delta^2\sqrt{1-\phi}\phi^2 + 4.5EcPr\eta^4\delta^2\sqrt{1-\phi}\phi^2 \\ -0.9EcPr\eta^6\delta^2\sqrt{1-\phi}\phi^2 + 1.5A_2\lambda(1-\phi)^3\phi^2 \\ -1.5A_2\lambda\eta^2(1-\phi)^3\phi^2). \end{cases} \quad (35)$$

Thus, the solution for the temperature profile is obtained as follows:

$$\theta(\eta) = \theta_0(\eta) + \theta_1(\eta) + \dots \quad (36)$$

Besides, the missing parameters  $f'(0), f'''(0)$  and  $\theta(0)$  do not contain in Eqs. (33) and (36). Therefore, such parameters are obtained numerically using RK4 method with shooting scheme as  $f'(0) = 1.41582016$ ,  $f'''(0) = -2.01604099$  and  $\theta(0) = 1.11883156$  when  $S = 1, Pr = 6.20, Ec = 0.010$ ,  $\phi = 0.020$  (Cu-water),  $\delta = 0.010$ ,  $N = 0.5$  and  $\lambda = 1.5$  respectively.

## 5. Results and discussion

The present study is all about the development of the mathematical model of the unsteady nanofluid flow squeezing through the parallel plates as shown in Fig. 1, where the significance influence of the thermal radiation and the heat generation/absorption parameters along with other relevant

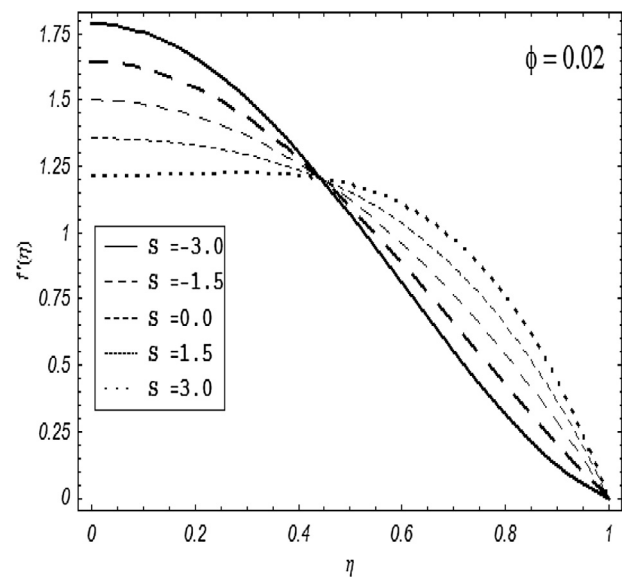
parameters on the flow and heat transfer features have been studied. However, the missing parameters  $f'(0), f'''(0)$  and  $\theta(0)$  have been computed using the numerical integration.

The present study is being validated with the results secured using fourth-order Runge-Kutta technique with shooting scheme, as shown in Fig. 2 and Table 1, respectively. The approximated values for both the velocity and the temperature profiles in Table 1, exhibit the mutual and excellent agreements between the two methods. Additionally, Table 2 displays the validation between the present results and the previously published results [2].

In Table 3 we found that the temperature gradient increases with the increase in the heat generation parameter. From Fig. 3, one can able to heed a satisfying agreement on the temperature profiles between the present results and the previously published results for Domairry and Hatami [42] obtained via DTM-Padé.

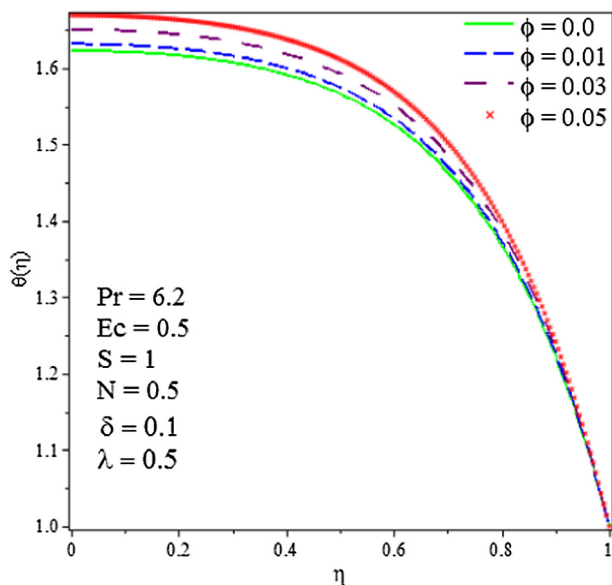
Moreover, Table 4 shows the effects of both Eckert number and Prandtl number on the temperature gradient, where it is noticed that the rate of heat transfer enhances significantly with the increase of both  $Ec$  and  $Pr$ , respectively. Table 5 depicts the effects of nanoparticles volume fraction  $\phi$  and squeeze integer  $S$  on velocity and temperature gradients with their corresponding skin friction and Nusselt number, where the velocity and temperature gradients with their corresponding skin friction and the Nusselt number have all been increased with the increase in the values of  $\phi$  and  $S$  significantly.

Fig. 4 displays the effects of squeeze integer on the velocity profile. The squeeze integer  $S$ , reports the motion of the plates,

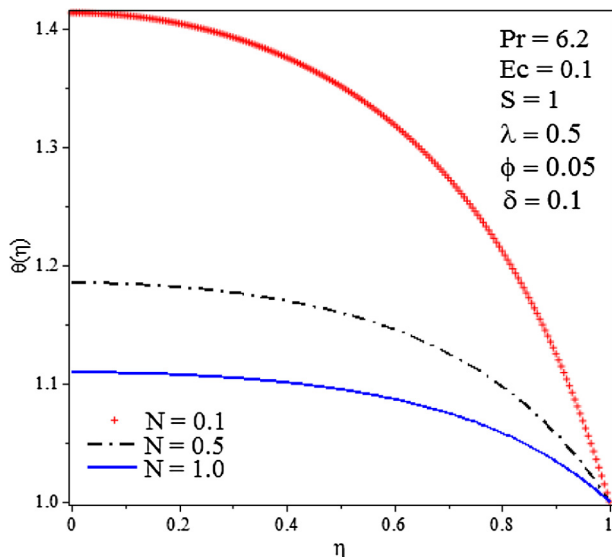
**Figure 4** Influence of squeeze number of nanofluid on the velocity profile (TiO<sub>2</sub>-water).

that is to say when  $S > 0$  it coincides with the plates moving separately, while  $S < 0$  coincides with the plates moving collectively, means the squeezing flow. Both positive and negative squeeze integers have distinct impacts on the velocity profile. The velocity increases with the increasing absolute value of squeeze integer when  $\eta < 0.5$ , while it reduces for  $\eta > 0.5$ .

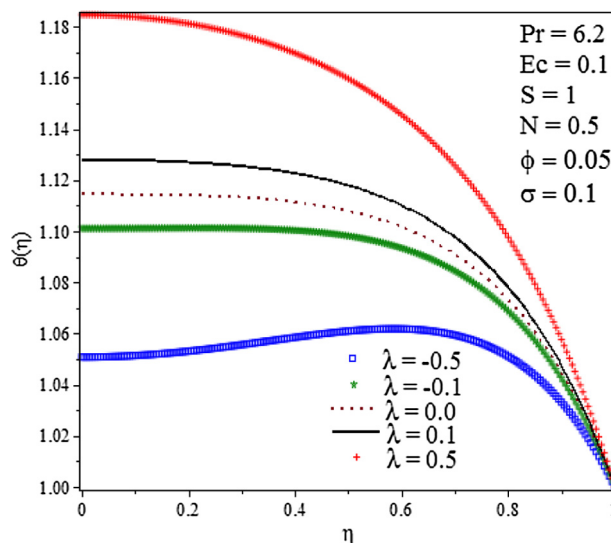
Intensifying the quantity of nano-sized particles into the conventional fluid, results to rise in thermal boundary layer thickness. However, it is well apparent that whenever the nano-sized particles are being increased into the fluid, the heat transfer is expected to have been enhanced, this is owing to their great thermal conductivity, and therefore the temperature profile is believed to have been reduced. But contrary to this hypothesis, the temperature notably increases due to the thermal radiation effect as depicted in Fig. 5. The thermal radiation effect is being depicted in Fig. 6, from which we noticed



**Figure 5** Influence of volume fraction of nanoparticle on the temperature profile (TiO<sub>2</sub>-water).



**Figure 6** Influence of thermal radiation on  $\theta(\eta)$  (TiO<sub>2</sub>-water).



**Figure 7** Heat source influence on the temperature profile (TiO<sub>2</sub>-water).

that, increasing the radiation parameter results to the decrease in the temperature profile.

It is heeded from Fig. 7 that the fluid temperature is physically rising in the case of heat generation ( $\lambda > 0$ ) and decreases in the case of heat absorption ( $\lambda < 0$ ) respectively.

## 6. Conclusions

In this study, HPM was applied on the transformed nonlinear ordinary differential equations to analyze the effects of both the thermal radiation and the heat generation/absorption on the approximate solution of unsteady squeezing nanofluid flow between two parallel plates. Moreover, the RK4 technique along with shooting scheme was used to testify the accuracy of our results. Thus, the desired and fascinate agreement between the analytical and numerical results was achieved. The impacts of all the salient parameters involved have been discussed and presented in tables and graphs respectively. Finally, based on the promising and attractive features of the results produced by HPM, the method is highly recommended for its capability of handling many nonlinear problems in the field of science and engineering.

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## References

- [1] M. Siddiqui, S. Irum, A.R. Ansari, Unsteady squeezing flow of a viscous MHD fluid between parallel plates, a solution using the homotopy perturbation method, *Math. Model. Anal.* 13 (4) (2008) 565–576.
- [2] M. Sheikholeslami, D.D. Ganji, H.R. Ashorynejad, Investigation of squeezing unsteady nanofluid flow using ADM, *Powder Technol.* 239 (2013) 259–265.

- [3] M. Sheikholeslami, D.D. Ganji, Heat transfer of Cu-water nanofluid flow between parallel plates, *Powder Technol.* 235 (2013) 873–879.
- [4] E.A. Hamza, The magneto hydrodynamic squeeze film, *J. Fluid Mech.* 19 (1964) 395.
- [5] E.A. Hamza, The magneto hydrodynamic effects on a fluid film squeezed between two rotating surfaces, *J. Phys. D: Appl. Phys.* 24 (1991) 547–554.
- [6] S. Bhattacharyya, A. Pal, Unsteady MHD squeezing flow between two parallel rotating discs, *Mech. Res. Commun.* 24 (6) (1997) 615–623.
- [7] J.H. He, Homotopy perturbation technique, *Comput. Methods Appl. Mech. Eng.* 178 (1999) 257–262.
- [8] J.H. He, A coupling method of homotopy technique and perturbation technique for nonlinear problems, *Int. J. Non-linear Mech.* 35 (2000) 37–43.
- [9] J.H. He, Homotopy perturbation method, a new non-linear analytical technique, *Appl. Math. Comput.* 135 (2003) 73–79.
- [10] J.H. He, Homotopy perturbation method for solving boundary value problems, *Phys. Lett. A* 350 (1–2) (2006) 87–88.
- [11] A.H. Nayfeh, *Introduction to Perturbation Techniques*, John Wiley & Sons, New York; Chichester, 1981.
- [12] A. Krylovas, Asymptotic method for approximation of resonant interaction of nonlinear multidimensional hyperbolic waves, *Math. Model. Anal.* 13 (1) (2008) 47–54.
- [13] A. Dib, A. Haiahem, B. Bou-said, Approximate analytical solution of squeezing unsteady nanofluid flow, *Powder Technol.* 269 (2015) 193–199.
- [14] O. Pourmehran, M. Rahimi-Gorji, M. Gorji-Bandpy, D.D. Ganji, Analytical investigation of squeezing unsteady nanofluid flow between parallel plates by LSM and CM, *Alexandria Eng. J.* 54 (2015) 17–26.
- [15] A.K. Gupta, S. Saha Ray, Numerical treatment for investigation of squeezing unsteady nanofluid flow between two parallel plates, *Powder Technol.* 279 (2015) 282–289.
- [16] U. Khana, N. Ahmed, M. Asadullah, S.T. Mohyud-din, Effects of viscous dissipation and slip velocity on two-dimensional and axisymmetric squeezing flow of Cu-water and Cu-kerosene nanofluids, *Propul. Power Res.* 4 (1) (2015) 40–49.
- [17] Analytical solution of unsteady GO-water nanofluid flow and heat transfer between two parallel moving plates, *Indian J. Chem. Technol.* 23 (2016) 47–52.
- [18] N. Hedayatil, A. Ramiar, Investigation of two phase unsteady nanofluid flow and heat transfer between moving parallel plates in the presence of the magnetic field using GM, *Trans. Phenom. Nano Micro Scales* 4 (1) (2016) 52–58.
- [19] T. Hayat, T. Muhammad, A. Qayyum, A. Alsaedi, M. Mustafa, On squeezing flow of nanofluid in the presence of magnetic field effects, *J. Mol. Liq.* 213 (2016) 179–185.
- [20] R. Kandasamy, I. Muhaimin, A.B. Khamis, R. Roslan, Unsteady Hiemenz flow of Cu-nanofluid over a porous wedge in the presence of thermal stratification due to solar energy radiation: lie group transformation, *Int. J. Therm. Sci.* 65 (2013) 196–205.
- [21] M. Mustafa, A. Mushtaq, T. Hayat, A. Alsaedi, Radiation effects in three-dimensional flow over a bi-directional exponentially stretching sheet, *J. Taiwan Inst. Chem. Eng.* 47 (2015) 43–49.
- [22] A.S. Dogonci, D.D. Ganji, Investigation of MHD nanofluid flow and heat transfer in a stretching/shrinking convergent/divergent channel considering thermal radiation, *J. Mol. Liq.* 220 (2016) 592–603.
- [23] A.S. Dogonci, D.D. Ganji, Thermal radiation effect on the nano-fluid buoyancy flow and heat transfer over a stretching sheet considering Brownian motion, *J. Mol. Liq.* 223 (2016) 521–527.
- [24] A.S. Dogonci, K. Divsalar, D.D. Ganji, Flow and heat transfer of MHD nanofluid between parallel plates in the presence of thermal radiation, *Comput. Methods Appl. Mech. Eng.* 310 (2016) 58–76.
- [25] A. Alsaedi, M. Awais, T. Hayat, Effects of heat generation/absorption on stagnation point flow of nanofluid over a surface with convective boundary conditions, *Commun. Nonlinear Sci. Numer. Simulat.* 17 (2012) 4210–4223.
- [26] M.A. Mansour, S.E. Ahmed, A numerical study on natural convection in porous media-filled an inclined triangular enclosure with heat sources using nanofluid in the presence of heat generation effect, *Eng. Sci. Technol. Int. J.* 18 (2015) 485–495.
- [27] A.H. Srinivasa, A.T. Eswara, Effect of internal heat generation or absorption on MHD free convection from an isothermal truncated cone, *Alexandria Eng. J.* 55 (2016) 1367–1373.
- [28] M.S. Abdel-wahed, E.M.A. Elbasheshy, T.G. Emam, Flow and heat transfer over a moving surface with non-linear velocity and variable thickness in a nanofluid in the presence of Brownian motion, *Appl. Math. Comput.* 254 (2015) 49–62.
- [29] M. Wahiduzzaman, Md.S. Khan, I. Karim, MHD convective stagnation flow of nanofluid over a shrinking surface with thermal radiation, heat generation and chemical reaction, *Proc. Eng.* 105 (2015) 398–405.
- [30] N.G. Rudraswamy, B.J. Gireesha, M.R. Krishnamurthy, Effect of internal heat generation/absorption and viscous dissipation on MHD flow and heat transfer of nanofluid with particle suspension over a stretching surface, *J. Nanofluids* 5 (6) (2016) 1000–1010.
- [31] M. Sheikholeslami, D.D. Ganji, Nanofluid hydrothermal behavior in existence of Lorentz forces considering Joule heating effect, *J. Mol. Liq.* 224 (2016) 526–537.
- [32] M. Sheikholeslami, D.D. Ganji, M.M. Rashidi, Magnetic field effect on unsteady nanofluid flow and heat transfer using Buongiorno model, *J. Magn. Magn. Mater.* 416 (2016) 164–173.
- [33] M. Sheikholeslami, D.D. Ganji, Nanofluid flow and heat transfer between parallel plates considering Brownian motion using DTM, *Comput. Methods Appl. Mech. Eng.* 283 (2015) 651–663.
- [34] M. Sheikholeslami, Ali J. Chamkha, Influence of Lorentz forces on nanofluid forced convection considering Marangoni convection, *J. Mol. Liq.* 225 (2017) 750–757.
- [35] G. Domairry, A. Aziz, Approximate analysis of MHD squeeze flow between two parallel disks with suction or injection by homotopy perturbation method, *Math. Probl. Eng.* 2009 (2009) 603916.
- [36] P. Bhattacharya, S.K. Saha, A. Yadav, P.E. Phelan, R.S. Prashar, Brownian dynamics simulation to determine the effective thermal conductivity of nanoparticles, *J. Appl. Phys.* 95 (2004) 6492–6494.
- [37] S. Rosseland, *Astrophysik und atom-theoretische Grundlagen*, Springer Verlag, Berlin, 1931, pp. 41–44.
- [38] N.S. Akbar, S. Nadeem, R.U.I. Haq, Z.H. Khan, Radiation effects on MHD stagnation point flow of nanofluid towards a stretching surface with convective boundary condition, *Chin. J. Aeronaut.* 26 (6) (2013) 1389–1397.
- [39] M. Kothandapani, J. Prakash, Effects of thermal radiation parameter and magnetic field on the peristaltic motion of Williamson nanofluids in a tapered asymmetric channel, *Int. J. Heat Mass Transf.* 81 (2015) 234–245.
- [40] M. Mustafa, T. Hayat, S. Obaidat, On heat and mass transfer in the unsteady squeezing flow between parallel plates, *Meccanica* 47 (2012) 1581–1589.
- [41] M. Sheikholeslami, H.R. Ashorynejad, D.D. Ganji, A. Yildirim, Homotopy perturbation method for three-dimensional problem of condensation film on inclined rotating disk, *Sci. Iran. B* 19 (3) (2012) 437–442.
- [42] G. Domairry, M. Hatami, Squeezing Cu-water nanofluid flow analysis between parallel plates by DTM-Padé method, *J. Mol. Liq.* 193 (2014) 37–44.